CHAPTER 1

SUCCESSIVE DIFFERENTIATION AND LEIBNITZ'S THEOREM

1.1 Introduction

Successive Differentiation is the process of differentiating a given function successively n times and the results of such differentiation are called successive derivatives. The higher order differential coefficients are of utmost importance in scientific and engineering applications.

Let $f(x)$ be a differentiable function and let its successive derivatives be denoted by $f'(x), f''(x), \dots, f^{(n)}(x)$.

Common notations of higher order Derivatives of $y = f(x)$

1st Derivative: $f'(x)$ or y' or y_1 or $\frac{dy}{dx}$ or Dy

2nd Derivative: $f''(x)$ or y'' or y_2 or $\frac{d^2y}{dx^2}$ or D^2y

$\vdots$

n^{th} Derivative: $f^{(n)}(x)$ or $y^{(n)}$ or y_n or $\frac{d^ny}{dx^n}$ or D^ny

1.2 Calculation of nth Derivatives

i. n^{th} Derivative of e^{ax}

Let $y = e^{ax}$

$$y_1 = ae^{ax}$$
$$y_2 = a^2e^{ax}$$
$$\vdots$$
$$y_n = a^n e^{ax}$$

ii. n^{th} Derivative of $(ax+b)^m$, m is a +ve integer greater than n

Let $y = (ax+b)^m$

$$y_1 = m\,a(ax+b)^{m-1}$$
$$y_2 = m(m-1)a^2(ax+b)^{m-2}$$
$$\vdots$$
$$y_n = m(m-1)\dots(m-n+1)a^n(ax+b)^{m-n}$$

Scanned with CamScanner

In a similar manner, we can find the fourth derivative, fifth derivative and, in gener derivative of y w. r. t. x by differentiating successively the given function y w. r. t. x f five times and n times.

Following notations are generally used for the successive derivatives of y w. r. t. x :

	First derivative	Second derivative	Third derivative	....	n^{th} derivative
or	y_1	y_2	y_3	....	y_n
or	$f'(x)$	$f''(x)$	$f'''(x)$	....	$f^{n}(x)$
or	$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	....	$\frac{d^ny}{dx^n}$
or	Dy	D^2y	D^3y	....	D^ny

Scanned with CamScanner

Scanned with CamScanner

Example 1 : If $\sqrt{x+y}+\sqrt{y-x}=c$, prove that $y_2=\frac{2}{c^2}$.

Solution : Given that $\sqrt{x+y}+\sqrt{y-x}=c$

Squaring both sides, we get

$$x+y+y-x+2\sqrt{y^2-x^2}=c^2$$

This $\Rightarrow 2\sqrt{y^2-x^2}=c^2-2y$

Again, squaring both sides, we get

$$4\left(y^2-x^2\right)=c^4-4c^2y+4y^2$$

This $\Rightarrow 4x^2-4c^2y+c^4=0$

Differentiating both sides w. r. t. x, we get

$$8x-4c^2y_1=0$$

This $\Rightarrow 2x-c^2y_1=0$

Again, differentiating both sides w. r. t. x, we get

$$2-c^2y_2=0$$

This $\Rightarrow y_2=\frac{2}{c^2}$.

Scanned with CamScanner

Scanned with CamScanner

Example 2 : If $y = a\cos(\log x) + b\sin(\log x)$, prove that $x^2y_2 + xy_1 + y = 0$.

Solution : Given that $y = a\cos(\log x) + b\sin(\log x)$.

Differentiating both sides w. r. t. x, we get

$$y_1 = -a\sin(\log x).\frac{1}{x} + b\cos(\log x).\frac{1}{x}$$

This $\Rightarrow xy_1 = -a\sin(\log x) + b\cos(\log x)$

Again, differentiating both sides w. r. t. x, we get

$$xy_2 + y_1 = -a\cos(\log x).\frac{1}{x} - b\sin(\log x).\frac{1}{x}$$

This $\Rightarrow x^2y_2 + xy_1 = -a\cos(\log x) - b\sin(\log x)$

$$= -(a\cos(\log x) + b\sin(\log x))$$

$$= -y \quad (\because y = a\cos(\log x) + b\sin(\log x) \text{ given})$$

This $\Rightarrow x^2y_2 + xy_1 + y = 0$.

Scanned with CamScanner

Scanned with CamScanner

Example 3 : If $y = e^{a\sin^{-1}x}$, prove that $(1-x^2)y_2 - xy_1 = a^2 y$.

Solution : Given that $y = e^{a\sin^{-1}x}$.

Differentiating both sides w. r. t. x, we get

$$y_1 = e^{a\sin^{-1}x}\left(a\frac{1}{\sqrt{1-x^2}}\right)$$

This $\Rightarrow y_1 = \dfrac{ay}{\sqrt{1-x^2}}$ $\quad \left(\because y = e^{a\sin^{-1}x} \text{ given}\right)$

$$\Rightarrow y_1\sqrt{1-x^2} = ay$$

Squaring both sides, we get

$$y_1^2(1-x^2) = a^2y^2$$

Again, differentiating both sides w. r. t. x, we get

$$2y_1y_2(1-x^2) - y_1^2(-2x) = a^2(2yy_1)$$

This $\Rightarrow 2y_1\left[(1-x^2)y_2 - xy_1\right] = 2y_1(a^2y)$

$$\Rightarrow (1-x^2)y_2 - xy_1 = a^2y.$$

Scanned with CamScanner

Scanned with CamScanner

Example 5 : Find n^{th} differential coefficient of $\frac{1}{(1+x)^2}$.

Solution : Let $y = \frac{1}{(1+x)^2}$

Differentiating both sides w. r. t. x successively, we get

$$y_1 = (-2)(1+x)^{-3}$$

$$y_2 = (-2)(-3)(1+x)^{-4}$$

$$y_3 = (-2)(-3)(-4)(1+x)^{-5}$$

.....

.....

Proceeding in this way, we get

$$y_n = (-2)(-3)(-4)\ldots\ldots\ldots\ldots(-(n+1))(1+x)^{-(n-2)}$$

i.e., $$y_n = \frac{(-1)^n (n+1)!}{(1+x)^{(n+2)}}.$$

Scanned with CamScanner

Scanned with CamScanner

$$= e^{3x}5^n \left[x^2 \sin\left(4x + n\tan^{-1}\frac{4}{3}\right) + \frac{2nx}{5}\sin\left(4x + (n-1)\tan^{-1}\frac{4}{3}\right) + \frac{n(n-1)}{25}\sin\left(4x + (n-2)\tan^{-1}\frac{4}{3}\right)\right]$$

Example 13 If $y = a\cos(\log x) + b\,sin(\log x)$, show that

$$x^2y_{n+2} + (2n+1)xy_{n+1} + n(n+1)y_n = 0$$

Solution: Here $y = a\cos(\log x) + b\sin(\log x)$

$$\Rightarrow y_1 = \frac{-a}{x}\sin(\log x) + \frac{b}{x}\cos(logx)$$

$$\Rightarrow xy_1 = -a\sin(logx) + b\,cos(logx)$$

Differentiating both sides w.r.t. x, we get

$$xy_2 + y_1 = -\frac{a}{x}\cos(logx) + \frac{-b}{x}\,sin(logx)$$

$$\Rightarrow x^2y_2 + xy_1 = -\{acos(logx) + b\,sin(logx)\}$$

$$= -y$$

$$\Rightarrow x^2y_2 + xy_1 + y = 0$$

Using Leibnitz's theorem, we get

$$(y_{n+2}x^2 + n_{C_1}y_{n+1}2x + n_{C_2}y_n.2) + (y_{n+1}x + n_{C_1}y_n.1) + y_n = 0$$

$$\Rightarrow y_{n+2}x^2 + y_{n+1}2nx + n(n-1)y_n + y_{n+1}x + ny_n + y_n = 0$$

$$\Rightarrow x^2y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

Example 14 If $y = \log(x + \sqrt{1+x^2})$

Prove that $(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0$

Solution: $y = \log(x + \sqrt{1+x^2}$

$$\Rightarrow y_1 = \frac{1}{x+\sqrt{1+x^2}}\left(1 + \frac{1}{2\sqrt{1+x^2}}2x\right) = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow (1+x^2){y_1}^2 = 1$$

Differentiating both sides w.r.t. x, we get

$$(1+x^2)2y_1y_2 + 2x{y_1}^2 = 0$$

$$\Rightarrow (1+x^2)y_2 + xy_1 = 0$$

Scanned with CamScanner

Using Leibnitz's theorem

$$[y_{n+2}(1+x^2) + n_{C_1}y_{n+1}2x + n_{C_2}y_n.2] + (y_{n+1}x + n_{C_1}y_n.1) = 0$$

$$\Rightarrow y_{n+2}(1+x^2) + y_{n+1}2nx + n(n-1)y_n + y_{n+1}x + ny_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0$$

Example 15 If $y = \sin(m\sin^{-1}x)$, show that

$(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2-m^2)y_n$. Also find $y_n(0)$

Solution: Here $y = \sin(m\sin^{-1}x)$ .....①

$$\Rightarrow y_1 = \frac{m}{\sqrt{1-x^2}}\cos(m\sin^{-1}x) \quad .....②$$

$$\Rightarrow (1-x^2){y_1}^2 = m^2\cos^2(m\sin^{-1}x)$$

$$\Rightarrow (1-x^2){y_1}^2 = m^2[1-\sin^2(m\sin^{-1}x)]$$

$$\Rightarrow (1-x^2){y_1}^2 = m^2(1-y^2).....③$$

$$\Rightarrow (1-x^2){y_1}^2 + m^2y^2 = m^2$$

Differentiating w.r.t. x, we get

$$(1-x^2)2y_1y_2 + {y_1}^2(-2x) + m^2 2yy_1 = 0$$

$$\Rightarrow (1-x^2)y_2 - xy_1 + m^2y = 0$$

Using Leibnitz's theorem, we get

$$[y_{n+2}(1-x^2) + n_{C_1}y_{n+1}(-2x) + n_{C_2}y_n(-2)] - (y_{n+1}x + n_{C_1}y_n1) + m^2y_n = 0$$

$$\Rightarrow y_{n+2}(1-x^2) - y_{n+1}2nx - n(n-1)y_n - (y_{n+1}x + ny_n) + m^2y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2-m^2)y_n.....④$$

Putting $x = 0$ in ①, ②and ③

$y(0) = 0, y_1(0) = m$ and $y_2(0) = 0$

Putting $x = 0$ in ④

$y_{n+2}(0) = (n^2-m^2)y_n(0)$

Putting $n = 1,2,3$ ... ... in the above equation, we get

$y_3(0) = (1^2-m^2)y_1(0)$

$=(1^2-m^2)m \quad \because y_1(0) = m$

Scanned with CamScanner